

	Physics-AI Fellow & Sansom Research Associate E-mail: af940@cam.ac.uk Website: alesfav.github.io LinkedIn: linkedin.com/in/alesfav X: x.com/alesfav	Centre for Mathematical Sciences University of Cambridge Cambridge, UK Cambridge-Infosys AI Centre London, UK
RESEARCH INTERESTS	Physics of learning/AI: generalization, data structure, compositionality, geometric priors, scaling laws, probabilistic graphical models, formal grammars, statistical mechanics. Foundation models: diffusion models, world models, multimodal models, post-training, model editing, model merging, interpretability, AI for scientific discovery.	
CURRENT AND PAST POSITIONS	University of Cambridge , Cambridge, UK Research Associate Inaugural <i>Physics-AI Fellow (independent postdoc)</i> in the Department of Applied Mathematics and Theoretical Physics (DAMTP) and <i>Sansom Research Associate</i> at Emmanuel College. Member of the <i>Cambridge-Infosys AI Centre</i> . Amazon Web Services Artificial Intelligence (AWS AI) , Santa Clara, California Applied Scientist Internship at AWS AI Labs on vision-language alignment of multimodal LLMs with the fundamental research team led by Prof. Stefano Soatto. EPFL , Lausanne, Switzerland Predoctoral Research Scholar <i>Master's valorization</i> research scholarship on the statistical physics of AI systems in the Institute of Physics. INRiM – Italian National Metrology Research Institute , Torino, Italy Research Intern Undergraduate internship on space-time quantum correlations in the Quantum Optics Laboratory led by Prof. Marco Genovese.	October 2025 to Present July to October 2023 November 2020 to April 2021 October 2017 to January 2018
EDUCATION	EPFL , Lausanne, Switzerland Ph.D., Physics, AI Advisors: Prof. Matthieu Wyart, Prof. Pascal Frossard. Dissertation: “The physics of data and tasks: Theories of locality and compositionality in deep learning”. G-Research EPFL PhD award in maths and data science (first prize). Sorbonne Université , Paris, France M.S., Fundamental Physics, Specialization in Complex Systems <i>Mention très bien</i> (highest honors). SISSA, ICTP, Politecnico di Torino , Trieste-Torino, Italy M.S., Physics of Complex Systems, Mathematical Modeling <i>110/110 cum laude</i> (highest honors). International Honors Track (competitive admission, 20 students per cohort). Thesis at EPFL: “Spectral analysis of infinitely-wide convolutional neural networks”. Politecnico di Torino , Torino, Italy B.S., Engineering Physics, Information Engineering Thesis: “Topological quantum computation with non-abelian anyons”.	2025 2020 2020 2018

AWARDS

- G-Research EPFL PhD prize in maths and data science (5,000 CHF), 2025.
- Recipient of *Physics-AI Fellowship*, DAMTP & Cambridge-Infosys AI Centre, 2025.
- Notable reviewer, ICLR, 2025.
- Dean's award for teaching excellence (1,000 CHF), EPFL, 2024.
- Top reviewer award, NeurIPS, 2024.
- Six-months *Master's valorization* research scholarship on statistical mechanics of deep learning (15,000 CHF), EPFL Institute of Physics, 2020.
- Merit-based scholarship for thesis abroad, Politecnico di Torino (2,500 EUR), 2020.
- Erasmus+ scholarship (3,000 EUR), Sorbonne Université, 2019.
- Fee reduction for high academic performance, Politecnico di Torino, 2019.
- Physics of complex systems international track fellowship, Politecnico di Torino, SISSA, ICTP (1,800 EUR), 2018.
- Top 200 engineering admission tests (8,000 applicants), Politecnico di Torino, 2014.

REFEREED PUBLICATIONS

Automatically-updated list on [Google Scholar](#).

Bibliographic metrics: 900+ citations, h-index 10+.

* *denotes co-first authorship.*

- [1] Cagnetta, F., Favero, A., Sclocchi, A. and Wyart, M., 2025. Scaling laws and representation learning in simple hierarchical languages: Transformers versus convolutional architectures. *Physical Review E*, 112(6), p.065312.
PRE Machine Learning Spotlight.
- [2] Wang, K., Qin, Y., Dimitriadis, N., Favero, A. and Frossard, P., 2025. MEMOIR: Lifelong Model Editing with Minimal Overwrite and Informed Retention for LLMs. To appear in *Advances in Neural Information Processing Systems (NeurIPS)*, 38.
- [3] Favero*, A., Sclocchi*, A., Cagnetta, F., Frossard, P. and Wyart, M., 2025. How compositional generalization and creativity improve as diffusion models are trained. *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, PMLR 267, pp.16286-16306.
Workshop version presented at the ICLR 2025 Workshop on Deep Generative Model in Machine Learning: Theory, Principle and Efficacy.
- [4] Sclocchi*, A., Favero*, A., Levi*, N. I. and Wyart, M., 2025. Probing the Latent Hierarchical Structure of Data via Diffusion Models. *International Conference on Learning Representations (ICLR)*.
Workshop version presented at the NeurIPS 2024 Workshop on Scientific Methods for Understanding Deep Learning. **Oral.**
Included in the 2025 special issue on the Statistical Physics aspects of Machine Learning, *Journal of Statistical Mechanics: Theory and Experiment*, 2025(8), p.084005.
- [5] Wang, K., Dimitriadis, N., Favero, A., Ortiz-Jimenez, G., Fleuret, F. and Frossard, P., 2025. LiNeS: Post-training Layer Scaling Prevents Forgetting and Enhances Model Merging. *International Conference on Learning Representations (ICLR)*.
- [6] Sclocchi, A., Favero, A. and Wyart, M., 2025. A Phase Transition in Diffusion Models Reveals the Hierarchical Nature of Data. *Proceedings of the National Academy of Sciences (PNAS)*, 122 (1), e2408799121.
- [7] Hazimeh*, A., Favero*, A. and Frossard, P., 2024. Task Addition and Weight Disentanglement in Closed-Vocabulary Models. *ICML 2024 Efficient Systems for Foundation Models Workshop*.
- [8] Cagnetta, F., Petrini, L., Tomasini, U.M., Favero, A. and Wyart, M., 2024. How Deep Neural Networks Learn Compositional Data: The Random Hierarchy Model. *Physical Review X*, 14(3), p.031001.
- [9] Favero, A., Zancato, L., Trager, M., Choudhary, S., Perera, P., Achille, A., Swaminathan, A. and Soatto, S., 2024. Multi-Modal Hallucination Control by Visual

Information Grounding. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp.14303-14312.
 Also presented at MMFM2: The 2nd Workshop on What is Next in Multimodal Foundation Models?, Seattle, WA, 2024.

- [10] Ortiz-Jimenez*, G., Favero*, A. and Frossard, P., 2023. Task Arithmetic in the Tangent Space: Improved Editing of Pre-Trained Models. Advances in Neural Information Processing Systems (NeurIPS), 36, pp.66727-66754.
Oral (top 0.54% of 12,000+ submissions).
- [11] Barak, B., Carrell, A., Favero, A., Li, W., Stephan, L. and Zlokapa, A., 2024. Computational complexity of deep learning: Fundamental limitations and empirical phenomena. Journal of Statistical Mechanics: Theory and Experiment, 2024(10), p.104008.
- [12] Cagnetta*, F., Favero*, A. and Wyart, M., 2023. What Can Be Learnt With Wide Convolutional Neural Networks?. Proceedings of the 40th International Conference on Machine Learning (ICML), PMLR 202, pp.3347-3379.
 Included in the 2024 special issue on the Statistical Physics aspects of Machine Learning, Journal of Statistical Mechanics: Theory and Experiment, 2024(10), p.104020.
- [13] Favero*, A., Cagnetta*, F. and Wyart, M., 2021. Locality defeats the curse of dimensionality in convolutional teacher-student scenarios. Advances in Neural Information Processing Systems (NeurIPS), 34, pp.9456-9467.
 Included in the 2022 special issue on the Statistical Physics aspects of Machine Learning, Journal of Statistical Mechanics: Theory and Experiment, 2022(11), p.114012.
- [14] Petrini, L., Favero, A., Geiger, M. and Wyart, M., 2021. Relative stability toward diffeomorphisms indicates performance in deep nets. Advances in Neural Information Processing Systems (NeurIPS), 34, pp.8727-8739.
 Included in the 2022 special issue on the Statistical Physics aspects of Machine Learning, Journal of Statistical Mechanics: Theory and Experiment, 2022(11), p.114013.

PRE-PRINTS

- [15] Favero*, A., Sclocchi*, A. and Wyart, M., 2025. Bigger Isn't Always Memorizing: Early Stopping Overparameterized Diffusion Models. arXiv preprint arXiv:2505.16959.
 Workshop version presented at the ICML 2025 Workshop on the Impact of Memorization on Trustworthy Foundation Models.
- [16] Abdelraheem*, A., Favero*, A., Bovet, G. and Frossard P., 2025. Backdoor Unlearning Through Linear Task Decomposition in Multimodal Models. arXiv preprint arXiv:2510.14845.
 Workshop version presented at the ICML 2025 Workshop on Machine Unlearning for Generative AI.

CONFERENCE
ABSTRACTS

- [17] Favero, A., Sclocchi, A., Cagnetta, F., Frossard, P. and Wyart, M., 2025. Compositional Generalization and Creativity in Language Diffusion Models. ACL 2025 Workshop on Structure-aware Large Language Models.
- [18] Favero, A., Cagnetta, F. and Wyart, M., 2023. Statistical Mechanics of Infinitely-Wide Convolutional Networks. Bulletin of the American Physical Society.
- [19] Petrini, L., Favero, A., Geiger, M. and Wyart, M., 2023. Diffeomorphisms invariance is a proxy of performance in deep neural networks. Bulletin of the American Physical Society.

SELECTED TALKS

- **International Centre for Theoretical Physics**, Youth in high-dimensions: Recent progress in machine learning, high-dimensional statistics and inference, Trieste, 2026.
- **International Conference on Machine Learning Physics**, Okinawa, 2026.
- **Bernoulli Center**, Fundamental Principles of Learning and Representation Workshop, Lausanne, 2026. *How combinatorial creativity emerges in diffusion models.*

- **INRIA Lyon (MALICE)**, 2025. *A competition between timescales: Dynamics of generalization vs. memorization in diffusion models.*
- **Johns Hopkins University**, Physics of Learning Seminars, 2025. *A competition between timescales: Dynamics of generalization vs. memorization in diffusion models.*
- **Stanford University HAI**, Physics of Learning and Neural Computation Workshop, 2025. *Geometry and disentanglement of tasks in the weight space of pre-trained models.*
- **Flatiron Institute Center for Computational Mathematics**, New York, 2025. *Probing and learning latent structure in data with diffusion models.*
- **University of Cambridge**, DAMTP Seminar, 2025. *Creativity in diffusion models: Insights from statistical physics and compositional grammars.*
- **Perimeter Institute**, Theory + AI: Theoretical Physics for AI, Waterloo, 2025. *Creativity by compositionality in generative diffusion models.* [\[Recording\]](#)
- **Johns Hopkins University Department of Physics & Astronomy**, 2025. *Creativity by compositionality in generative diffusion models.*
- **IBM Research**, IBM Accelerated Discovery Seminar, Zurich, 2024. *Task arithmetic in the tangent space of pre-trained models.*
- **EPFL Center for Intelligent Systems**, 2023. *Task arithmetic in the tangent space: Improved editing of pre-trained models.*
- **Conference on Neural Information Processing Systems**, New Orleans, 2023. *Task arithmetic in the tangent space: Improved editing of pre-trained models.*
- **Amazon AI Labs**, 2023. *Task arithmetic in the tangent space of pre-trained models.*
- **MIT Center for Biological and Computational Learning**, 2023. *Deep convolutional networks in kernel regimes: invariances, locality, and compositionality.*
- **NYU Center for Data Science**, 2023. *Generalization properties of deep convolutional networks in kernel regimes.*
- **American Physical Society March Meeting**, Statistical Physics Meets Machine Learning, Las Vegas, 2023. *Statistical mechanics of infinitely wide convolutional networks.*
- **EPFL Institute of Physics**, Seminars in Physics of Bio/Complex Systems, 2023. *Symmetry, locality, and hierarchy in artificial neural networks.*
- **Rice University**, Workshop on the Theory of Overparameterized ML, 2022. *Locality defeats the curse of dimensionality in convolutional teacher-student scenarios.*
- **Flatiron Institute**, Center for Computational Neuroscience, New York, 2024. *Hierarchies and compositionality in diffusion models.*
- **Oxford Department of Statistics**, Workshop on Robustness in LLMs, Oxford, 2024. *Multi-modal hallucination control by visual information grounding.*
- **Princeton University ORFE Department**, Princeton, 2022. *How wide convolutional neural networks learn hierarchical tasks.*
- **Simons Foundation**, Simons Collaboration on Cracking the Glass Problem Meeting, New York, 2022. *Spatial locality and translational invariance in machine learning.*

SELECTED
POSTERS

MEETINGS AND SCHOOLS	• Mathematics of machine learning, Italian National Institute for Advanced Mathematics (INdAM – Istituto Nazionale di Alta Matematica), Cortona, 2024 (<i>invited</i>). • Analytical connectionism summer school, Flatiron Center for Computational Neuroscience, New York, 2024. • Machine learning theory summer school, Princeton University, Princeton, 2022. • Statistical physics and machine learning summer school, Les Houches School of Physics, Les Houches, 2022. • Youth in high-dimensions: machine learning, high-dimensional statistics, and inference for the new generation, ICTP, 2021. • The hitchhiker’s guide: machine learning for condensed matter (4 days), 2021, ICTP. • Spring college in physics of complex systems (1 week, interrupted due to the pandemic), 2020, ICTP, Trieste.
TEACHING	University of Cambridge, Cambridge, UK Supervisions for the MPhil in Data Intensive Science. Fall 2025 to present EPFL, Lausanne, Switzerland Teaching assistant (<i>2024 Dean’s award for teaching excellence</i>) Fall 2021 to present • PHYS-316 Statistical Physics II: Phase Transitions and Critical Phenomena (Spring 2023, Spring 2024). • PHYS-467 Machine Learning for Physicists (Fall 2021, Fall 2022, Fall 2023). • PHYS-421 Physics Projects I: Statistical Mechanics of Deep Learning (Fall 2021). Guest lecturer at CS-625 Transfer Learning and Meta-Learning (Spring 2024).
ADVISING AND MENTORING	(Co-) supervised 16 student research projects, including 11 Master’s theses, 3 Master’s internships, and 2 Ph.D. semester rotation projects across Physics, Data Science, and Computer Science programs.
REVIEWER	Reviewer/referee for 20+ papers per year in machine learning and physics. • Advances in Neural Information Processing Systems (NeurIPS). <i>2024 Top Reviewer</i>. • International Conference on Learning Representations (ICLR). <i>2025 Notable Reviewer</i>. • International Conference on Machine Learning (ICML). • IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). • Transactions on Machine Learning Research (TMLR). • Physical Review Journals.
ACADEMIC SERVICE	• ELLIS (European Lab for Learning & Intelligent Systems) PhD Recruiting Committee, Evaluator (a.y. 2024-25).
MEDIA & PRESS	• Physics Magazine, “Viewing neural networks through a statistical-physics lens”, Feb 2026. (Regarding PRE 2025 publication). [Article] • EPFL News, “A hidden hierarchy in AI image generation”, Mar 2025. (Regarding PNAS 2025 publication). [Article]
SOFTWARE	• Programming. Python, C, C++, UNIX shell scripting. • Scientific and ML Libraries. NumPy, Matplotlib, scikit-learn, PyTorch, JAX, HF Transformers. • HPC. SLURM, Docker, K8s, Amazon Elastic Compute Cloud (EC2).
CERTIFICATES	IELTS Academic 8.5/9 (CEFR C2).